

SAFEGUARDING CHIP SECURITY: Challenges And Countermeasures



Strategies such as enhanced testing, side-channel analysis, and security hardening can mitigate these threats, but challenges persist



VASUKI SHANKAR
is Senior Software
Engineer, NVIDIA

In semiconductor manufacturing, preserving the integrity and security of electronic chips emerges as a top priority. These electronic chips play an indispensable role in powering everyday gadgets and devices that individuals depend on, such as smartphones and laptops. Nonetheless, within this high-stakes arena, a pervasive menace known as hardware Trojans casts a substantial shadow over chip security. The ensuing discourse will explore the nature of hardware Trojans, their potential methods of implementation, and, perhaps most crucially, the strategies that can be employed to counteract them, ensuring the safeguarding of our digital lifestyles.

Understanding hardware Trojans

Hardware Trojans represent a class of malevolent functionalities surreptitiously incorporated into the Register Transistor Logic (RTL) netlist of a semiconductor chip before its actual fabrication. This nefarious tampering is typically orchestrated by malicious actors who manage to gain unauthorised access to the chip's design files during the manufacturing process. What makes hardware Trojans particularly insidious is their capacity to remain covert and undetected by end-users and customers, lurking in the chip's architecture until activated by a covert and unique sequence of actions known exclusively